

Meg Montañez

Davenport

Blockchain Health Systems Engineer

Web3 Architect · Applied Researcher

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[americaoutloud.news/author/](https://americaoutloud.news/author/meg-montanez-davenport)

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Blockchain & Web3

SolidityHardhatRemix IDE ERC-20/721Brave WalletEIP-6963 window.ethereumDIDs / SSI

IPFSFilecoinMerkle DAGs ZK Proofslibp2pBlockscout

Engineering & Data

Next.jsPostgreSQL 17Python FlaskStreamlitsystemd REST APIsVercelSSH Ed25519 Git

CI/CD Agile/ScrumTableau

Healthcare & Informatics

HL7FHIRROMOP HIPAAEHR Mapping Clinical Data ModelsOrthomolecular

Organizations

INCOSE — Systems Engineering

AMIA — Medical Informatics

HIMSS — Health IT

ISC² — Cybersecurity

PDMA — Product Management

ISOM — Orthomolecular Medicine

NANP — Nutrition Professionals

Certifications

Certified Blockchain Healthcare Professional (CBHP) — Blockchain Council · Apr 2025

Certified Smart Contract AuditorBlockchain Council · Jan 2026

Health Informatics — Johns HopkinsMay 2025

Web3 & Blockchain — INSEADApr 2025

CISA — PacktAug 2025

SSI Master — DizmeIDMar 2025

Medical Neuroscience — Duke SOMApr 2025

Health Informatics on FHIR — Georgia TechMar 2025

Business Analyst — IBM Jun 2025

Data Analytics · UX Design · PM — Google Jun 2025

IPFS · Filecoin · libp2p · Merkle DAGs IPFS / Filecoin Foundation / ProtoSchool · 2025

NC Accident & Health Insurance Producer Pre-licensing completed Apr 2026, exam scheduled

National Provider Identifier (NPI): 1497696264 Naturopath (175F00000X), NPPES, 2026

Summary

I design and deploy decentralized infrastructure for sovereign data governance — using regulated behavioral health as a deliberate proving ground for architectures that scale across complex domains. As a long-term healthcare practitioner with experience operating within HIPAA regulatory frameworks as a licensed clinical provider, I built within compliance frameworks before architecting beyond them — and FSL's infrastructure operates outside HIPAA scope by architectural design, holding no PHI. As founder of Future Systems Lab, I move from concept to production: on-chain deployments, EIP-191 wallet-gated consent architecture, and infrastructure designed to operate outside HIPAA scope by holding no PHI — a stronger architectural position than compliance, because the regulation doesn't apply when the system never holds the protected data. I think in systems, across disciplines, at the intersection of governance and user experience. Eight Ethereum smart contracts deployed and verified on Sepolia — every FSL system is in production. Pursuing a Doctor of Engineering to formalize applied research methodology and advance systems leadership at the intersection of blockchain, AI, and decentralized governance.

Engineering Experience

Lead Engineer — SovereignLedger Sovereign Health Governance Infrastructure

2026 — Present

Sepolia: [0x4afA577fA914068451e0Aa97b61F23960f02aCc4](#)

- Engineered a full PostgreSQL session governance schema spanning participants, sovereign guides, sessions, attestations, documents, audit logs, consent grants, and compliance log tables.
- Built Phase 3 SMS-based approval-gated automation with case-insensitive approve/deny/defer, 4-hour timeout, and max-defer logic — deployed to production VPS via systemd.

- Architected SovereignLedger as a standalone blockchain governance layer for session governance, attestation, and participant-controlled record verification — integrates with any payment pipeline (Crypto → XRP → Sovereign Navigation) and deployable independently of any single platform.

Practice Operations Lead — Riccobene Associates

2019 — 2025

70+ office practice network, North Carolina

- Led operational scaling across 70+ office practice network, producing 64% per-provider production growth (\$380,710 → \$624,640 average per provider) through team coordination, workflow optimization, and cross-location process standardization.
- Designed and deployed two-tier production-share compensation model adopted company-wide — 27% standard tier (network-wide rollout) and 33% architect tier (system designer compensation).
- Per-provider share of total office production grew from 30.4% to 38.1% — system redesign enhanced overall office economics rather than redistributing existing revenue.
- Recruited directly by CEO for C-suite engagement; maintained multi-year engagement while concurrently founding Future Systems Lab.

Open-Source Case Study Lead — AlchemistForge / mental-health-on-chain

2026 — Present

- Scaffolded mental-health-on-chain as a five-entry public smart contract awareness series; deployed 24/7 campaign orchestration across Dev.to, Twitter, Discord, GitHub, and Telegram.
- Recorded on-chain case study as Participant #1 with Transmuted and Celebrated events confirmed at wallet [0xf22cbf25deEeA36FFF828BAf73CCb049345eF248](https://etherscan.io/address/0xf22cbf25deEeA36FFF828BAf73CCb049345eF248).

Founder & Systems Architect — Future Systems Lab

2023 — Present

github.com/Future-Systems-Lab/Future-Systems-Lab-profile

- Deployed and verified eight Ethereum smart contracts on Sepolia testnet: [HypnoNeuroToken](#), [EHTv2](#), [MindMasteryNFT](#), [SovereignLedger](#), [AlchemistForge](#), [BenevolenceFund](#), [SovereignAchievement](#), and [HNT v2](#).
- Architected a 17-agent Master Command Center spanning Audit, Smart Contract, Backend Build, Frontend/UI, Compliance, Security, Testing/QA, and Research agents with full dropdown panel orchestration.

- Deployed production VPS automation stack (clawdbot) on OpenClaw VPS: Flask approval server, Telegram bot, Streamlit analytics dashboard, and two systemd services running continuously.
- Engineered EncryptAI — a 19-step canonical monorepo (frontend/, backend/, labs/, contracts/, game/) with Brave Wallet authentication via window.ethereum + EIP-191, PostgreSQL 17, and blockchain writes via viem — [Step 2 deployed as Secure Health Login](#).
- Established SSH-authenticated signed commits, GitHub org governance, and Rights Reserved licensing policy across all FSL repositories.

Healthcare Data Architect — HypnoNeuro Mental Wellness Platform

2023 — Present

HypnoNeuro · EncryptHealth · SovereignLedger · NeuroBalance Watch — modular systems engineered to operate independently or as an integrated ecosystem

- Architected and deployed a tier-based therapeutic game framework — currently operating with three flagship demos (one per level: L1 Theta Wave/GABA, L2 Biochemical/Serotonin, L3 Original Frequency/Endocannabinoid) demonstrating the full participant flow: wallet gating, HNT token rewards, on-chain achievement minting, and NFT-based tier access. Architecture supports expansion to full catalog and roadmap targets VR-mediated avatar circle integration.
- Architected dual-chain healthcare session anchoring — every session cryptographically committed to both Ethereum Sepolia and XRPL testnet, generating a tamper-evident audit trail for sovereign health records.
- Built ISOM-aligned 4-tier provider credentialing system (Platinum/Gold/Silver/Verified) with automated license verification via Verifiable.com and AI-assisted credential review using Claude Vision.
- Implemented Walsh Protocol AI recommendation engine — biotype classification across 5 neurotransmitter systems (Trp, Tyr, Cortisol, Dopamine, Epi/NorEpi) with plant-based nutrient protocol generation.
- Deployed avatar-based anonymous session rooms via Daily.co WebRTC — wallet-connected, provider-facilitated video sessions with sessionStorage-persisted character customization and SovereignLedger billing anchor.
- Designed white-label operator licensing model with three commercial tiers (Starter \$497/mo, Growth \$1,497/mo, Enterprise custom) — full registration flow, Telegram-notified applications pipeline, and operator config abstraction layer.

- Completed 35-step canonical build across 7 sprints — all five platforms live on Vercel and interconnected: HypnoNeuro, EncryptHealth, SovereignLedger, NeuroBalance, Provider Portal, and FSL Command Center.
- **Sprint 1–3:** Core platform buildout — therapeutic game framework, Brave Wallet + XRPL dual-wallet auth, PostgreSQL schema, FHIR R4 export, XRPL session anchoring, NFT gating, HNT token rewards, provider portal backend.
- **Sprint 4–5:** Avatar Session Rooms (Daily.co WebRTC), AlchemistForge HNT bridge, endocannabinoid baseline ML scorer, COA Lookup, Support Circles, Walsh Protocol recommendation engine, wellness scoring, XRPL wallet connect.
- **Sprint 6:** XRPL wallet polish (GemWallet/Crossmark/Xaman/manual), demo mode with watermark and seeded leaderboard, biosensor infrastructure spec (5 target hardware partners), FSL Command Center on Vercel, Telegram operator notification pipeline.
- **Sprint 7 (current):** Intake form Walsh Protocol biotype pre-screening, win screen confetti + NFT progress tracking + share flow, support circle categories and HNT bonus, ecosystem health monitoring (cron + Telegram alerts), OSS portfolio formalization.
- Planned hardware integration: NeuroBalance biosensor layer targeting Nutromics (sweat biomarkers), Polar (HRV), OpenBCI (EEG), Emotiv, and Dexcom (CGM) — full API spec and device schema documented.

Founder & Product Developer — SunCodes Holistic Health

2018 — 2024

- Developed a seed-to-shelf CBD product line in an unregulated market — managed full product lifecycle including formulation, third-party COA testing, GMP documentation, labeling, barcode integration, and retail packaging compliance.
- Led full renovation and interior redesign of a historic downtown smoke shop, converting it into a branded therapeutic wellness center — managed contractor relationships, spatial design, and retail buildout from concept to opening.
- Founded and operated a clinical wellness platform applying naturopathic psychology and orthomolecular frameworks to mental health and trauma recovery, with digital compliance tracking and personalized client protocol systems.

Open Source Contributions

Open Source Contributor — Healthcare & Blockchain

Jul 2025 — Present

Self-Directed / Independent

- [DIF / Presentation-Exchange #486](#) — OIDC4VP healthcare consent flow example · Open, awaiting named review by Daniel Buchner (csuwildcat), Kim Hamilton Duffy (kimdhamilton), Brent Zundel, and rado0x54 · Foundation for SovereignLedger's consent architecture and EIP-191 wallet authentication
- [OpenZeppelin Contracts #6414](#) — ReentrancyGuard documentation clarification · Open · FSL contracts inherit from OpenZeppelin audited base contracts
- [Hyperledger Fabric #5419](#) — Build and runtime environment notes · Open · Permissioned ledger patterns for FSL provider credentialing
- [Ethereum / EIPs #11409](#) — Markdown formatting fix to README's EIP document structure section (Abstract / Motivation / Specification / Rationale / Security Considerations) · Open, in active review · Referenced in EIP Editor Office Hours #93 (Mar 17, 2026) and #94 (Mar 24, 2026); all 9 automated checks passed · Improves rendering consistency for the constitutional document of the EIP standards process
- [Uniswap v3-core #1077](#) — Invariant test note for pool math · Open · Reference for future HNT/ETH liquidity pool design
- [Balancer v2 #2648](#) — Pool math reference documentation · Open · HNT liquidity pool and AlchemistForge bridge architecture reference
- [BigchainDB #2763](#) — Python quickstart documentation · Open · Immutable data patterns informing NeuroBalance biosensor anchoring
- [OpenMRS / FHIR2 #594](#) — Quickstart pointer for healthcare interoperability · Closed · Same FHIR2 module stack as EncryptHealth FHIR R4 export

Applied projects:

- [Secure Health Login](#) — practitioner-facing smart wallet authentication app implementing EIP-712, DID/OIDC, and sovereignty-first access control — live on Vercel

Publications

INSURANCE & CLAIMS GOVERNANCE

"Ending the Insurance Verification Nightmare"

America Out Loud · March 1, 2026 · americaoutloud.news

Proposed blockchain governance as a systemic solution to insurance fraud — duplicate claims, identity misuse, phantom billing — through decentralized eligibility verification and immutable audit trails.

MENTAL HEALTH & BEHAVIORAL WELLNESS

"Reprogram Your Limiting Beliefs on the Blockchain While Learning Web3"

America Out Loud · April 2026 · americaoutloud.news

Introduced AlchemistForge — Jungian shadow integration permanently recorded on Ethereum.
Aligns with Ghasemzadeh Lab behavioral health intervention and digital health systems research.

SECURITY, PRIVACY & CONSENT ARCHITECTURE

"Blockchain Consent Transforms Patient Data Control"

America Out Loud · March 2026 · americaoutloud.news

Examined how smart contract-governed consent flows resolve systemic failures of centralized health data management. Aligns with Ahn/SEFCOM identity management and access control research.

BLOCKCHAIN & WEB3 HEALTHCARE INFRASTRUCTURE

"Stryker Cyberattack and Blockchain in Healthcare"

America Out Loud · March 2026 · americaoutloud.news

Analyzed the Stryker Corporation cybersecurity incident and proposed decentralized infrastructure as a systemic alternative to centralized healthcare IT architectures.

HEALTH INFORMATICS & DATA SOVEREIGNTY

"Why Your Medical Record Belongs in Your Digital Wallet"

America Out Loud · April 2026 · americaoutloud.news

Argued that sovereign wallet-based health data ownership eliminates institutional gatekeeping. Aligns with Boscovic Lab blockchain healthcare data sovereignty research.

Full archive (weekly contributor, 2024–present): americaoutloud.news/author/meg-montanez-davenport

Research

EIP-191 Consent Architecture for Sovereign Behavioral Health Data Governance — Lead Author

Manuscript in preparation; target journal: Blockchain in Healthcare Today (BHTY)

Establishes the EIP-191 wallet-signature consent pattern as a generalizable architecture for sovereign data governance, with regulated behavioral health as the deliberate stress-test domain. Documents the implementation deployed in SovereignLedger v2 (Sepolia: [0x4afA577fA914068451e0Aa97b61F23960f02aCc4](#)).

Education

BS, Management Information Systems
CSU Global — Regionally Accredited (HLC)
Conferred July 2026
Doctor of Naturopathic Psychology
Kingdom College of Natural Health
Conferred

Entity & Intellectual Property

Future Systems Lab LLC (Wyoming) — EIN 42-2050630
USPTO Trademark "FUTURE SYSTEMS LAB" — Serial No. 99533250, Class 42 (filed Dec 2025; Office Action response filed Apr 2026, examiner review pending)
National Provider Identifier (NPI): 1497696264 — Naturopath (175F00000X), NPPES, 2026

Proposed Applied Research

Decentralized Sovereign Data Governance: Engineering a privacy-first, blockchain-secured session attestation system for sovereign data governance — advancing SovereignLedger from testnet deployment into a rigorous applied research framework examining smart contract-governed consent architecture operating outside HIPAA scope by holding no PHI, with behavioral health as the deliberate proving ground for the most regulated domain. | [SovereignLedger on Blockscout](#)